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Resource Optimization Algorithm for 5G Core Network

Integrating NFV and SDN Technologies

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Abstract: The growth in network demand has driven the development of new network technologies. However, traditional network architecture cannot meet the huge traffic of transportation and different business needs. To address this issue, a specific network service function chain is formed based on the network function virtualization. Dynamic resource awareness algorithms are introduced to construct an adaptive migration model based on network function virtualization. Based on the Multi-Armed Bandit (MAB) algorithm, a dynamic routing model based on MAB is constructed by using a greedy algorithm to search for random actions. When the nodes were 200 and 500, the migration costs of the adaptive migration model based on network function virtualization were 1000 and 3000, respectively. The average migration was 350 and 900 respectively, while destination nodes' average resource occupancy rates were 52% and 58%, respectively. When the path failure rates were 4% and 20%, the algorithm's safe path rates were 96.25% and 92.75%. For fixed and mobile nodes, the link load rate of the dynamic routing model based on the MAB algorithm was low and the load growth was relatively stable. This dynamic routing model's link delay is significantly less than the Dijkstra algorithm. These two models can maximize server resource utilization, reduce cost consumption, and achieve maximum overall benefits.

Keywords: Network function virtualization; Network service function chain; Dynamic resource perception; Adaptive migration; Multi-armed bandit

1. Introduction

As technologies such as cloud computing and the Internet of Things (IoT) continue to evolve, the integration of software defined network (SDN) and network function virtualization (NFV) technologies has become increasingly promising in the market. Conventional methods frequently implement fixed resource allocation strategies to fulfill particular service quality expectations. However, this fixed configuration is unable to adapt effectively to fluctuating traffic loads over time [1]. In a dynamic network environment, it is necessary to be able to adjust resource allocation schemes in real time to meet end-to-end latency requirements. Traditional methods often cannot often make timely adjustments in this regard. Meanwhile, traditional methods still face challenges in solving network resource optimization problems, such as complex problem transformations and difficult solutions [2]. Therefore, the research proposes the optimization algorithm of 5G core network resources using NFV and SDN technologies. The innovation of the research lies in the fact that although VNF-SFC and SDN have been widely studied, the study introduces dynamic resource awareness and Adaptive Migration (AM) strategies based on traditional methods. It does so to solve the problems of uneven resource allocation and high migration costs in large-scale network environments that have been identified in existing methods. Compared to static migration methods, the proposed AM model can dynamically adjust based on the real-time status of network nodes. This enables a notable reduction in migration costs and an enhancement in network resource utilization. Despite the extensive use of the MAB algorithm in this domain, research has

sought to enhance its efficacy in network path selection by refining the algorithmic parameters.

This study consists of four parts in total. Firstly, the research on SDN and NFV technology by industry scholars is introduced. Secondly, AM model is built based on the NFV and Dynamic routing model is built based on the Multi-Armed Bandit (MAB) routing algorithm. Next, this study tests the performance of NFV-based AM and MAB-based dynamic routing models. Finally, a summary of the entire article is provided and the shortcomings of this study are pointed out. The notation table is shown in Table 1.

Table 1 Notation table

Symbol	Meaning
α	The node calculation
β	The node storage state
γ	The network resources overload
$N_{w_j}^{core}$	The total computing resources owned by node
n_i^{core}	The resources required for node computing
n_i^{mem}	The storage resources required by node
$N_{w_j}^{mem}$	All storage resources owned by node
c_i	The network resources required by nodes
$C_{j,k}$	The link bandwidth between nodes
α_k	The overload situation of node k class resources
x_k	The proportion of k resources in total resources
x_{ik}	The proportion of migration consumption to total resources
K	The total resource occupancy of a node
b_{ik}	The virtual nf being migrated to the node
φ_0	The original bandwidth
φ_{jk+}	Additional bandwidth
φ_{max}	The maximum bandwidth
f_n	Network SFC
N	The set of all unmigrated nodes
T_{ki}	The constraint threshold of the migration node during migration
τ_k	Node resources' minimum utilization rate
φ_k	Node resources' maximum utilization rate
γ_i	Each indicator's booking definition weight
T_{ki}	Migration node's constraint threshold during migration
T_{i+}	Rational solution's occupancy rate
φ_+	Rational solution's additional bandwidth
T_{i-}	Negative ideal solution's occupancy rate
φ_-	Negative ideal solution's additional bandwidth
$load_{ij}$	Link's bandwidth utilization rate
$x_{ij}^k \cdot d_{ij}$	Path delay's cumulative multiplication
d_{ij}	Latency

s_{t-1}	The state obtained by agent at time t
a_t	Executing an action
s_t	The updated status
A	The set of actions
S	The set of networks
R	The reward function
α_1	Link bandwidth
α_2	Link delay
α_3	The adjustment factor for data packet loss rate
BW_{ij}	A real-time reward for link bandwidth
$delay_{ij}$	The proportion of link latency to total link latency
$loss_{ij}$	The proportion of link data packet loss rate to total link packet loss rate
$\pi_{i,j}^{s,t}$	The optimal strategy
M	The time spent searching for SDN controller
N_i	The total strategy for exploring node S to destination node t
M'	The time spent searching for SDN controller
N'_i	The total strategy from exploration node S to destination node t

2. Related works

SDN is a new type of network architecture that can achieve more precise network control, and some scholars have conducted relevant research on SDN. W Rafique et al. found that complex infrastructure, security vulnerabilities, and newly developed technologies in IoT have become challenging issues. To solve this problem, SDN and edge computing ecosystems have been extensively investigated and studied. The requirements and standardization work for integrating these different architectures are explained. There are some loopholes in the edge computing ecosystem. Based on these vulnerabilities, examples are analyzed in the experiment and vulnerabilities are classified [3]. Shaji N S et al. proposed an intelligent intrusion detection system for detecting distributed denial of service attacks in response to security threats in SDN. Three ensemble machine learning models were developed in the study, including support vector classifiers, random forests, and logistic regression, for the purpose of performing binary and multi-classification tasks. The results showed that the accuracy of multi-class random forest and logistic regression was 99.54%, which improved detection accuracy and sensitivity [4]. Alhaj AN et al. proposed a robust security architecture to address the security issues of SDN. This architecture consisted of multiple modules, each addressing specific security challenges. A multi-layer security algorithm was developed in the research to mitigate DDoS attacks, and a real-time algorithm was designed to handle ARP attacks, especially ARP request replay DoS attacks. The results indicated that this architecture could effectively mitigate critical security attacks such as DDoS [5]. K R Das et al. believed that there were issues such as high latency, heterogeneity, and packet loss in low-power wireless personal area networks. To address this issue, an edge SDN architecture was constructed by combining SDN with existing IoT structures. To reduce heterogeneity, a hybrid edge switch has been developed to facilitate communication between SDN and low-power wireless personal area networks. This method could ensure the optimal routing of data packets and achieve effective communication between devices [6]. To enhance the flexibility and certainty of industrial control networks, Z Pang et al. combined time

sensitive networks and SDNs to construct time sensitive SDNs. Moreover, a new flow scheduling Generative model was proposed to ensure that there will be no frame loss in the network update process. This method achieved zero frame loss without additional overhead, saved 40% of execution time when bandwidth utilization was below 50%, and improved schedulability [7].

NFV enables the conversion of network functions (NFs) that were previously reliant on hardware implementation into software. This facilitates the rapid deployment of NFs, elastic expansion, and the automation of network management functions. In actual network architecture, NFV can achieve maximum efficiency, and some scholars have conducted relevant research on this. To make the virtual network have better performance, Z Yang et al. proposed a new efficient virtual network automatic embedding algorithm by combining deep reinforcement learning (RL) with a graph convolution based neural network. Moreover, the RL framework and the newly designed multi-objective reward function were introduced into the training. Research has shown that compared with existing solutions, this method improved information acceptance rate by 39.6% and had strong robustness [8]. Biallach H et al. proposed a VNF reconfiguration solution to address the issue of network slicing reconfiguration in 5G networks. Two precise integer linear programming models were defined in the study and two approximate solutions were proposed. One of which is based on column generation method. Through numerical testing and comparison, the performance and quality of the algorithm have been verified, providing an effective solution [9]. Diaz G et al. proposed a solution for integrating virtualization during network deployment to efficiently deploy new services in rapidly changing digital environments. They focused on the impact of virtualization on network deployment, emphasizing the importance of flexibility, adaptability, and dynamism for network as a service. By using SDN and NFV, this study provided support for designing new network architectures [10]. Y Huang et al. believed that cloud data centers provided convenient services for an increasing number of applications, but the energy consumption and service quality issues in cloud service centers needed to be addressed. To address this issue, NFV was introduced and a cloud data center optimization allocation scheme based on the customer needs was proposed. Compared with existing methods, this method had good performance in terms of total energy consumption, memory consumption, the number of physical machines used, and the number of service protocol violations [11].

In previous studies, SDN and NFV have received widespread attention and research, respectively. SDN represents a novel approach to network architecture, offering enhanced precision in network control and improved utilization and response speed of network resources through the separation of the control plane and data plane. However, although these studies demonstrate the potential of SDN, there are still limitations in solving large-scale network resource optimization and path selection problems. In terms of NFV, research mainly focuses on how to virtualize traditional network functions and achieve efficient resource utilization through flexible deployment and automated management. However, these methods still face issues of uneven resource distribution and high migration costs when dealing with dynamic resource allocation in large-scale network environments. In the research on the combination of SDN and NFV, existing work has mainly focused on the integration and optimization of network architecture. Nevertheless, the majority of these tasks remain at the level of architectural design and functional implementation, exhibiting a paucity of in-depth exploration of dynamic resource perception and migration strategies. Based on existing literature, a dynamic resource-aware AM and a dynamic routing model based on the MAB algorithm are proposed. Unlike existing methods,

this study focuses on addressing the issues of uneven resource allocation and high migration costs in large-scale network environments. The AM model optimizes the utilization efficiency of network resources by dynamically adjusting migration strategies through real-time monitoring of node resource occupancy. Meanwhile, the MAB routing model combines real-time decision-making mechanisms in dynamic environments, effectively reducing network latency and link load.

3. Construction of 5G Core Network Resource Optimization Model

To meet network consumption cost and functional adaptation in 5G core network, this chapter is divided into two parts to construct a network architecture. The first part constructs AM based on the NFV to reduce network cost consumption. The second part builds a dynamic routing model based on the MAB routing algorithm to achieve the balance between network resources and dynamic routing.

3.1 AM Construction on the Foundation of NFV

SDN is a new type of network architecture that has the characteristics of classification control plane and data plane [12, 13]. Usually, the control plane of an SDN is managed by one or more controllers, and the data plane is composed of a series of switches or routers. The controller is responsible for managing the flow table, routing table, and other information in the network. Additionally, the controller is tasked with the provision of corresponding instructions and policies for the data plane [14]. NFV differs from traditional network hardware in that it does not necessitate dedicated hardware for implementation. Instead, NFV employs virtualization technology to transform NF that was originally designed for hardware implementation into software. In the context of NFV, the resource usage of network nodes is constantly changing due to the dynamic nature of resource occupancy. The main purpose of migration is to ensure the efficient operation of network functions when resource usage exceeds a certain threshold [15]. Therefore, dynamic resource-aware algorithms are used to monitor the real-time resource occupancy of nodes and migrate VNF when necessary, in order to achieve load balancing and optimal resource allocation. The designation "adaptive" is applied to this migration strategy because it is capable of adapting to the actual state of the network in real time. It is able to select the optimal migration node, minimize migration costs, optimize network performance, and avoid unnecessary migrations. Fig. 1 is a schematic diagram of SDN and NFV fusion.

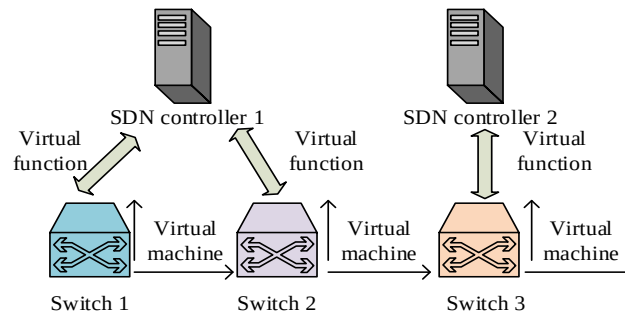


Fig. 1 Schematic Diagram of SDN and NFV Technological Convergence

During NFV migration, a specific network service function chain (SFC) is formed to meet the needs of network services. Due to the end-to-end performance constraints of SFC during NFV migration, bandwidth increases exponentially, increasing network costs [16]. Therefore, it is necessary to minimize the cost of NFV migration and achieve dynamic load balancing of the entire network. In a migration network, the amount of resources contained by each node has a

corresponding constraint threshold. When the resource occupancy exceeds this constraint threshold, it will cause NFV migration. Fig. 2 is a schematic diagram of NFV migration.

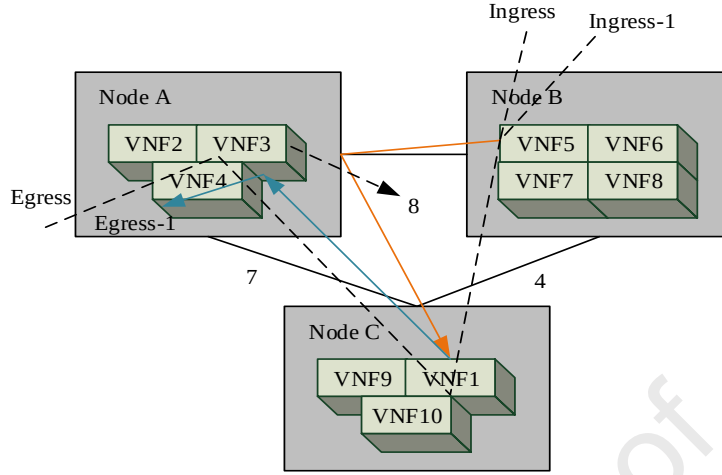


Fig. 2 Schematic Diagram of NFV Migration

In the context of NFV, considering the dynamic nature of resource usage, the main motivation for adopting dynamic resource-aware algorithms is to monitor the resource occupancy status of network nodes in real-time to ensure efficient network operation [17]. According to the occupancy of resources, node computing's occupancy rates, storage, and network can be calculated in equation (1).

$$\begin{cases} \alpha = \frac{n_i^{core}}{N_{w_j}^{core}} \\ \beta = \frac{n_i^{mem}}{N_{w_j}^{mem}} \\ \gamma = \frac{c_i}{\sum_{k \neq j, k \in V^{Node}} C_{j,k}} \end{cases} \quad (1)$$

In equation (1), α stands for the node calculation, β stands for the node storage state, and γ stands for the network resources overload. $N_{w_j}^{core}$ stands for the total computing resources owned by node, while n_i^{core} stands for the resources required for node computing. n_i^{mem} stands for the storage resources required by node, while $N_{w_j}^{mem}$ stands for all storage resources owned by node. c_i stands for the network resources required by nodes and $C_{j,k}$ stands for the link bandwidth between nodes. By using DRP algorithm, the migration index of NFV can be obtained in equation (2).

$$\theta_i = \sum_K \left\{ \alpha_k \frac{1}{1 - x_k} + (1 - \alpha_k)(1 - x_k) \right\} x_{ik}, \quad K = (Core, Mem, Com) \quad (2)$$

In equation (2), α_k stands for the overload situation of node k class resources. x_k stands for the proportion of k resources in total resources. x_{ik} stands for the proportion of migration consumption to total resources. K stands for the total resource occupancy of a node. When the resources in the node exceed the constraint threshold, θ is greater than 1. When the resource does not exceed the constraint threshold, θ is less than 1. Based on the calculated migration index, resource types with higher migration indices are selected for priority migration. Equation (3) stands for the constraint conditions for migration on nodes.

$$\text{subject to: } \sum a_{ij} = 1, a_{ij} \in \{0,1\} \quad (3)$$

In equation (3), a_{ij} stands for that the virtual network function (v_i) of NFV must and can only be deployed on one node. According to SFC performance constraints where NFV to be migrated is located, nodes that meet the conditions are selected for migration. Equation (4) is the constraint requirement formula.

$$a_{ij} \cdot b_{ik} \cdot \Delta \varphi_{jk+} + \varphi_0 < \varphi_{\max}, v_i \in f_n \quad (4)$$

In equation (4), b_{ik} stands for the virtual NF being migrated to the node. φ_0 stands for the original bandwidth. φ_{jk+} stands for additional bandwidth. $\varphi_{\max}$ stands for the maximum bandwidth. f_n stands for network SFC. By using equation (2), the nodes with large resource surplus to be migrated are selected, and the multi-objective decision-making TOSIS algorithm is introduced to measure the migration index [18]. By calculating the positive ideal solution, the best migration solution among the nodes to be migrated is selected. Equation (5) stands for rational solution's occupancy rate.

$$T_{i+} = \min\{T_{ki} | k \in N_3\}, i \in \{1, 2, 3\} \quad (5)$$

In equation (5), N stands for the set of all unmigrated nodes. T_{ki} stands for the constraint threshold of the migration node during migration. The maximum additional bandwidth of all selected nodes needs to meet the conditions in equation (6).

$$\varphi_+ = \min\{\tau_k | k \in N_3\} \quad (6)$$

In equation (6), τ_k stands for node resources' minimum utilization rate. The maximum utilization rate of resources is taken as the negative ideal solution in equation (7).

$$T_{i-} = \max\{T_{ki} | k \in N_3\}, i \in \{1, 2, 3\} \quad (7)$$

Equation (7) can represent the worst migration plan among the nodes to be migrated. After DRP analysis, the additional bandwidth of the negative ideal solution satisfies the conditional formula in equation (8).

$$\varphi_- = \max\{\varphi_k | k \in N_3\} \quad (8)$$

In equation (8), φ_k stands for node resources' maximum utilization rate. According to equation (9), the Euclidean distance between the node and the rational solution is calculated.

$$\theta_{ok+} = \sqrt{\sum_{i=1}^3 [\gamma_i (T_{ki} - T_{i+})]^2 + [\gamma_4 (\varphi_k - \varphi_+)]^2} \quad (9)$$

In equation (9), γ_i stands for each indicator's booking definition weight. T_{ki} stands for migration node's constraint threshold during migration. T_{i+} stands for rational solution's occupancy rate. φ_+ stands for rational solution's additional bandwidth. Equation (10) is node and negative ideal solution's Euclidean distance.

$$\theta_{ok-} = \sqrt{\sum_{i=1}^3 [\gamma_i (T_{ki} - T_{i-})]^2 + [\gamma_4 (\varphi_k - \varphi_-)]^2} \quad (10)$$

In equation (10), T_{i-} stands for negative ideal solution's occupancy rate. φ_- stands for negative ideal solution's additional bandwidth. Equation (11) stands for the closeness of positive and negative ideal solutions.

$$\theta_{ok} = \frac{\theta_{ok-}}{\theta_{ok-} + \theta_{ok+}} \quad (11)$$

In equation (11), θ_{ok} is larger, the node is closer to the ideal solution. Choosing the node with the highest θ_{ok} value as the migration target can maximize resource utilization and save network consumption costs. The AM model is based on NFV and aims to solve the problem of uneven resource allocation in traditional network environments. With the growth of network traffic, dynamic resource allocation has become the key to ensuring service quality. Fig. 3 shows the AM flowchart.

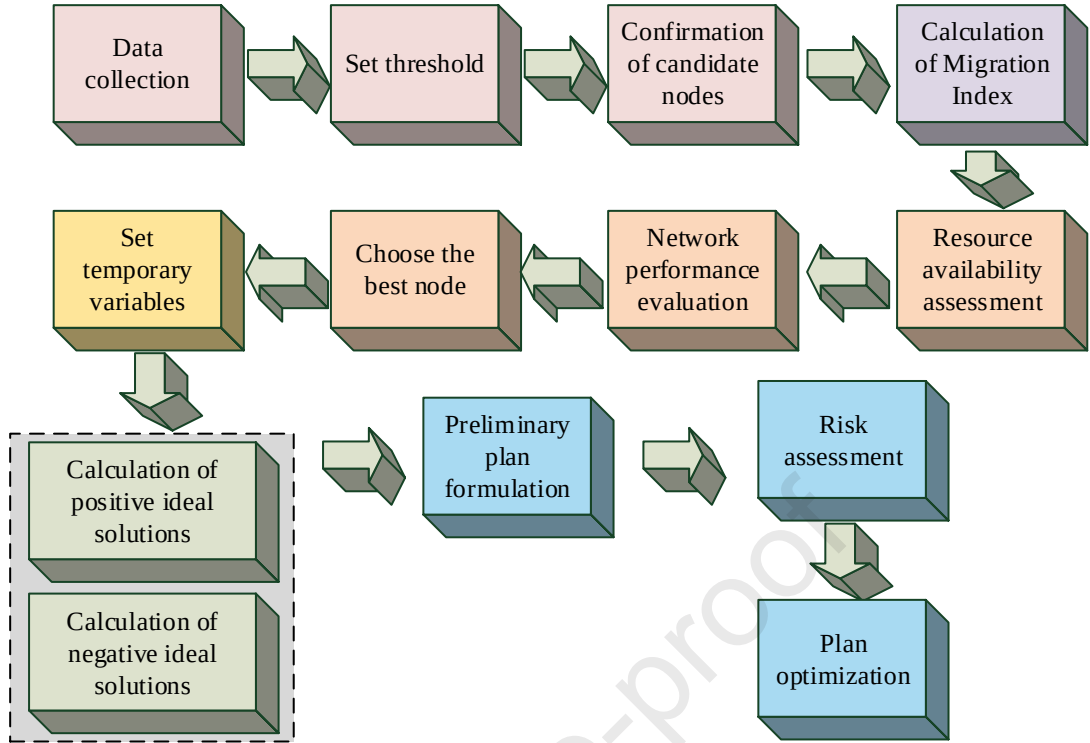


Fig. 3 Adaptive Migration Model

In Fig. 3, the real-time monitoring of resource occupancy of each node is achieved through the dynamic resource awareness algorithm. When the resource usage of a node exceeds the set threshold, the model will identify the virtual network functions that need to be migrated. The migration decision is based on the collected resource occupancy data, and a multi-objective decision algorithm is used to evaluate the transferable nodes. The model calculates the migration index for each node to select the optimal migration target and maximize resource utilization. During the migration process, the model dynamically adjusts the network load to achieve load balancing across the entire network.

3.2 Construction of Dynamic routing Model on the foundation of MAB

The reason for choosing the MAB algorithm in designing dynamic routing models is that it can make real-time decisions in uncertain environments. The MAB algorithm helps the network select the optimal transmission path by continuously exploring and utilizing the benefits of network paths to balance load and reduce latency. This decision-making mechanism can effectively improve the allocation of network traffic in multiple paths, thereby enhancing overall network performance. SDN can select different routing path plans within a limited time, achieving maximum overall efficiency [19, 20]. Fig. 4 shows an SDN routing model on the foundation of MAB.

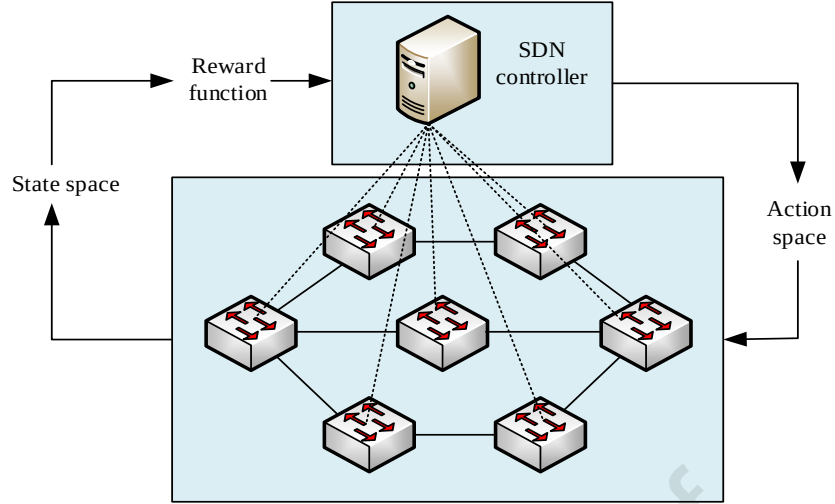


Fig. 4 SDN routing model on the foundation of multi-arm slot machines

In SDN, there are n nodes connected by undirected links. When selecting links, consideration will be given to the network's business characteristics, which mainly include source nodes, destination nodes, bandwidth, latency, and data packet loss rate [21]. While ensuring the normal operation of network business functions, the load balancing of links has been improved as much as possible. Equation (12) stands for link's bandwidth utilization rate.

$$\lambda \max_{(i,j) \in E} load_{ij} + (1-\lambda) \sum_{k \in E} \sum_{(i,j) \in E} x_{ij}^k \cdot d_{ij} \quad (12)$$

In equation (12), $load_{ij}$ stands for link's bandwidth utilization rate. $x_{ij}^k \cdot d_{ij}$ stands for path delay's cumulative multiplication. x_{ij}^k indicates that the route has been transmitted. d_{ij} stands for latency. The link bandwidth has been classified. When the transmission traffic reaches more than 80% of the maximum bandwidth value, it indicates excessive link load. When the transmission traffic reaches 30% to 80% of the maximum bandwidth value, it indicates the link load. When transmission traffic reaches less than 30% of the maximum bandwidth value, it indicates that the link is slightly overloaded. The MAB contains an SDN controller, which allows the intelligent agent to take actions on the foundation of the received network environment status information [22]. Actions are routing choices in network links, and the choices made are related to environment state. In MAB, random selection strategies, Softmax, and upper confidence bound (UCB) algorithms are commonly used to solve problems [23]. Therefore, Epsilon green (EG) is used for random action search. Each path scheme found in this model is represented as an arm, and equation (13) is each arm's action value function.

$$\begin{cases} Q(s_t, a_t) = \alpha * R(s_t, a_t) + (1-\alpha) * \sum_{k=1}^{t-1} R(s_k, \alpha_k) \\ = \alpha * R(s_t, a_t) + (1-\alpha) * Q(s_{t-1}, a_{t-1}) \end{cases} \quad (13)$$

In equation (13), $R(s_t, a_t)$ stands for the reward at time t . s_{t-1} stands for the state obtained by agent at time t . a_t stands for executing an action. s_t stands for the updated status.

The relationship between state, action, and reward function can be represented by equation (14).

$$S \times A \rightarrow R \quad (14)$$

In equation (14), A stands for the set of actions, S stands for the set of networks, and R stands for the reward function. After taking a series of actions, the feedback loop will upload the reward function $R(s_t, a_t)$ and update the action value function $Q(s_t, a_t)$ and matrix Q . Fig. 5 shows the RL model.

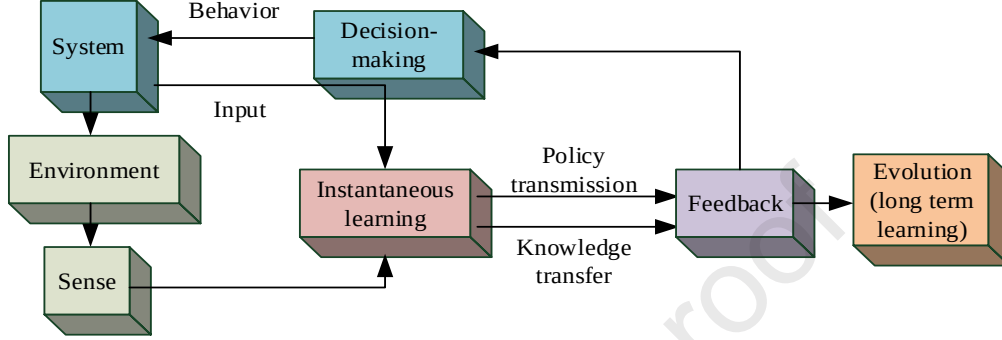


Fig. 5 RL Model

In Fig. 5, agent can send information to the environment on the foundation of status, actions, and reward values. Then environment provides feedback after receiving information. This feedback return serves as next feedback's experience and the next action decision [24]. After the above operation, agent will select the optimal strategy on the foundation of the expected reward value. Equation (15) is the optimal strategy formula.

$$Q^*(s_t, a_t) = \max_{a \in A} Q(s_t, a) \quad (15)$$

In equation (15), $Q(s_t, a)$ stands for the defined expected reward. When transmitting traffic on the network path, there may be traffic overload. Therefore, when selecting a path, it is necessary to consider the overall traffic restriction problem in network. The setting of reward function is related to network business requirements, network conditions, and SDN controller's actions. For network services that are not sensitive to data loss rates, reward function's data loss rate parameter will be prioritized and reduced [25]. Therefore, Sigmoid function is set as SDN dynamic routing model's reward function in equation (16).

$$R_{i \rightarrow j} = R(i, j | s_t, a_t) = -\text{cost} + \alpha_1 BW_{ij} - \alpha_2 \text{delay}_{ij} - \alpha_3 \text{loss}_{ij} \quad (16)$$

In equation (16), α_1 stands for link bandwidth. α_2 stands for link delay. α_3 stands for the adjustment factor for data packet loss rate. BW_{ij} stands for a real-time reward for link bandwidth. delay_{ij} stands for the proportion of link latency to total link latency. loss_{ij} stands for the proportion of link data packet loss rate to total link packet loss rate. SDN controller's reward function is related to network business requirements, network operation strategies, and controller actions [26]. The link performance selected by MAB is better, the reward function value is larger. When planning SDN controller routing, SDN controller will perform link search on the

global network to obtain the best action. Equation (17) stands for the effective action set.

$$\Pi_{useful} = \{\pi_{i,j}^{s,t}(t_i)\}, i = 1, 2, \dots, M; j = 1, \dots, N_i \quad (17)$$

In equation (17), $\pi_{i,j}^{s,t}$ stands for the optimal strategy. M stands for the time spent searching for SDN controller. N_i stands for the total strategy for exploring node s to destination node t . Equation (18) is a invalid action set.

$$\Pi_{useless} = \{\pi_{i,j}^{s,Vt_i}(t'_i)\}, i = 1, 2, \dots, M'; j = 1, \dots, N'_i \quad (18)$$

In equation (18), M' stands for the time spent searching for SDN controller. N'_i stands for the total strategy from exploration node s to destination node t . In MAB, pursuing the highest existing reward and constantly exploring new and higher rewards are contradictory. Therefore, it is not only necessary to maximize the highest existing reward's utilization, but also to solve for higher rewards. Introducing MAB in SDN can achieve the optimal path selection method. Fig. 6 shows the MAB based dynamic routing model.

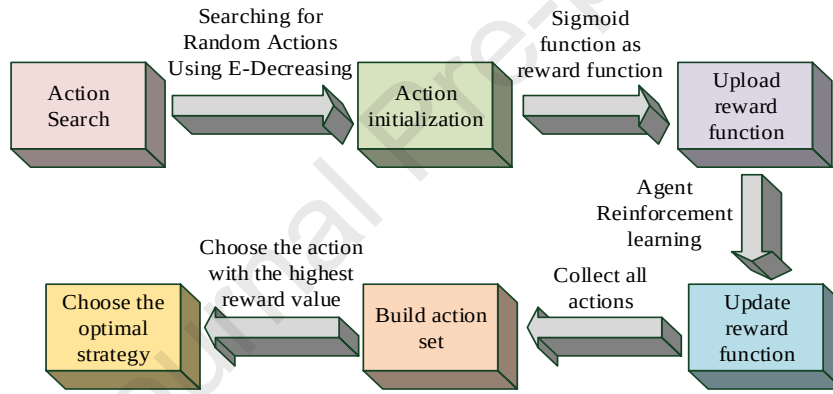


Fig. 6 Flow chart of Dynamic Routing Model Based on the MAB Algorithm

4. Performance Testing of 5G Core Network Resource Optimization Model

A 5G core network resource optimization model is constructed in the study, and this chapter will be divided into two parts to test the research model. The first is NFV-based AM and the second is MAB-based dynamic routing model. Based on the test results, two models of network resource optimization are analyzed.

4.1 AM analysis and Testing Based on NFV

When conducting model analysis testing, the simulation settings and detailed parameters studied are shown in Table 2.

Table 2 Simulation Settings and Detailed Parameters

Parameter Name	Value
Number of Network Nodes	100, 200, 500
Number of Links	500
Bandwidth per Link	5 Gbps
Number of Network Functions (NF)	1-5
Number of Service Function Chains (SFC)	5
Migration Trigger Threshold	80%

Maximum Path Failure Rate	4%, 10%, 20%
Learning Rate (MAB Algorithm)	0.3, 0.6, 0.9
Discount Factor (MAB Algorithm)	0.3, 0.6, 0.9
Initial Resource Allocation	Random
Simulation Duration	10 hours
Traffic Type	Random Distribution, Uniform Distribution
Link Load Rate	30%, 50%, 75%
Latency Measurement Interval	100 ms

The parameter settings used in the simulation are based on typical configurations in existing 5G networks and SDN to ensure high reliability of the simulation results. The settings of bandwidth and link load rate are based on data from actual 5G networks and are used to verify the effectiveness of the algorithm under high load conditions. This ensures that the simulated settings are consistent with the ideal conditions of real operation. The simulation first initializes the network environment and parameter settings, then sets the traffic model and configures algorithm parameters such as multi-arm slot machines. The simulation experiment runs by monitoring node resources and path selection performance, recording various performance data, and finally analyzing and visualizing the results to evaluate the effectiveness of the algorithm. During testing, the research method will be abbreviated as NFV-SDN optimization algorithm and compared with Fixed weight algorithm, PageRank TrustRank (PRT) algorithm, and Simple instant algorithm. Fig. 7 shows a comparison curve of the total migration cost.

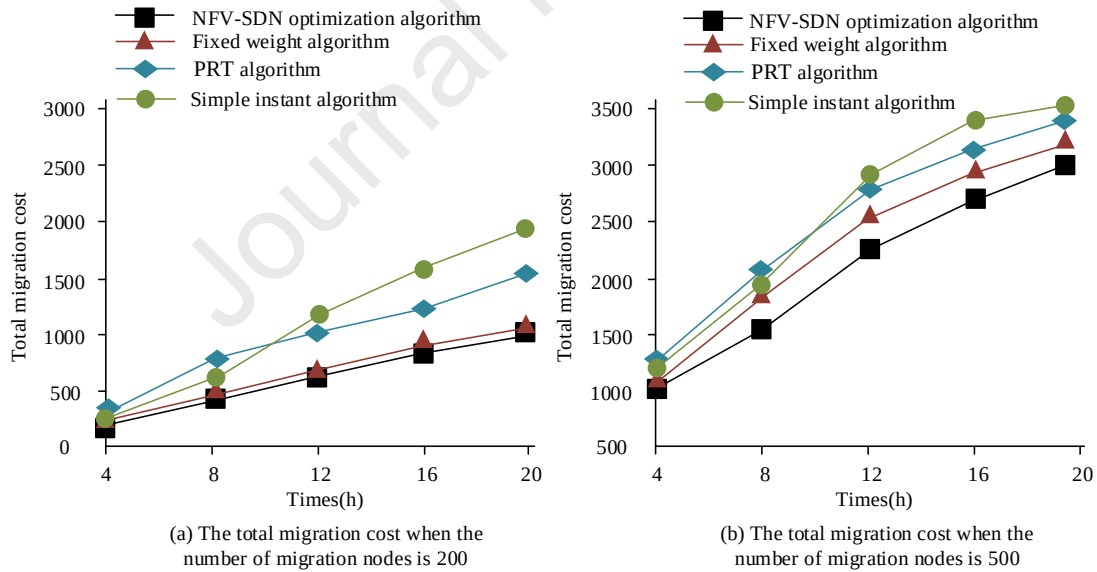


Fig. 7 Comparison curve of total migration cost

In Fig. 7, when node is 200, four algorithms' migration costs are significantly lower than when node is 500. Total migration cost Indicates the total resource consumption generated during the migration of virtual network functions, including bandwidth usage, computing resource reassignment, storage resource adjustment, and migration. The lower the total migration cost, the better the migration strategy performance in terms of resource utilization efficiency, load balancing capability, and network stability. The migration cost directly reflects the optimization ability of the algorithm in dynamic resource allocation, and the lower migration cost means less network fluctuation and higher service continuity, which is crucial for the high reliability and low

latency requirements of the 5G core network. In Fig. 7 (a), as testing time increases, four algorithms' migration costs increase slightly. At the initial testing stage, four algorithms' total migration costs are basically the same. As testing time increases, simple instant algorithm's migration cost exceeds that of the other three algorithms. At this point, the migration cost of NFV-SDN optimization algorithm is 1000. In Fig. 7 (b), this network migration's node number is 500, and all four algorithms' migration costs have significantly increased. Especially for simple instant algorithm, its migration costs are the highest. At this point, the migration cost of NFV-SDN optimization algorithm is 3000. From these two figures, NFV-SDN optimization algorithm has the lowest migration cost when network migration. Four algorithms are compared for average migration within the specified time in Fig. 8.

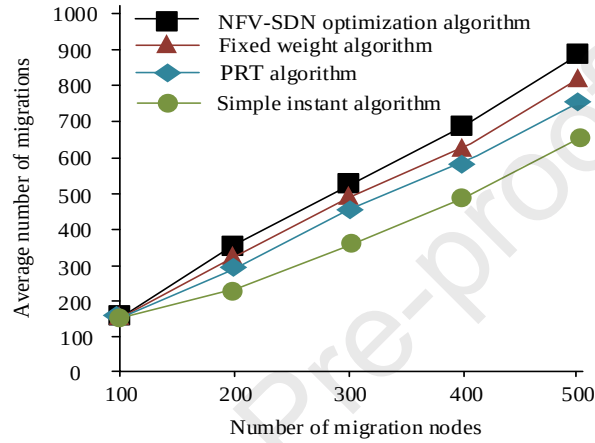


Fig. 8 Comparison curve of average number of migrations

From Fig. 8, as nodes increase, all four algorithms' average migrations improve. When migration node is 100, four algorithms' average migrations are basically the same, with a quantity of 150. When migration node is above 200, the average number of NFV-SDN optimization algorithm starts to exceed other three algorithms. At this point, the average migration of NFV-SDN optimization algorithm is 350. When node is 500, the average migration of NFV-SDN optimization algorithm is the highest, with a quantity of 900. Therefore, AM algorithm based on NFV outperforms other algorithms in decision-making during migration. Four models are tested 10 times, with nodes of 200 and 500. The migration destination node's resource occupancy is compared in Fig. 9.

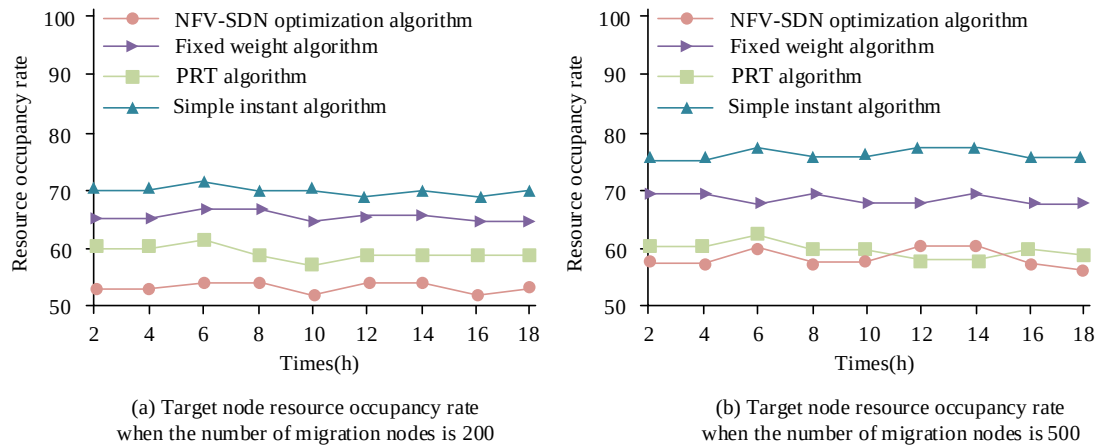


Fig. 9 Resource occupancy of migration destination nodes for different algorithms

From Fig. 9 (a), fixed weight algorithm's average target node resource occupancy rate is 65%.

Simple instant algorithm's average target node resource occupancy rate is 70%. PRT algorithm's average target node resource occupancy rate is 58%. The average target node resource occupancy rate in NFV-SDN optimization algorithm is 52%. From Fig. 9 (b), fixed weight algorithm's average target node resource occupancy rate is 68%. Simple instant algorithm's average target node resource occupancy rate is 75%. PRT algorithm's average target node resource occupancy rate is 60%. The average target node resource occupancy rate in NFV-SDN optimization algorithm is 58%. The increase of nodes will lead to an increase in target node resource occupancy. However, the target node resource occupancy of NFV-SDN optimization algorithm is the least compared to other four algorithms. Therefore, introducing multi-objective decision-making algorithms into migration algorithms is effective and can effectively reduce resource consumption. In particular, the increase in the average number of migrations reflects the ability of the algorithm to actively adjust resource allocation in a dynamic network environment. When the node size increases, resource overload becomes more frequent. Higher migration times indicate that the NFV-SDN optimization algorithm can identify and migrate key virtual network functions in real time through the multi-objective decision algorithm, thereby avoiding local resource congestion and improving the overall resource utilization. In contrast, other algorithms adopt static strategies or inefficient decision-making mechanisms, insufficient migration times or redundancy, resulting in a significant increase in resource usage of target nodes. It shows that higher migration times are directly related to better decision quality, that is, load balancing is achieved through dynamic migration, which ultimately reduces resource waste and improves network stability. It is further proved that the research method has better decision-making effect than other algorithms in the process of migration. Shortest Path routing (SPR), node disjoint multi-path shortest path routing (NDMP-SPR), efficient routing (ER), and node disjoint multi-path efficient routing (NDMP-ER) algorithms are selected for comparison with NFV-SDN optimization algorithm. Table 3 shows the safe path rate.

Table 3 Comparison of Path Security Rates of Different Algorithms

Failure rate(%)	NFV-SDN optimization algorithm	SPR	NDMP-SP R	ER	NDMP-E R
4	0.9625	0.7516	0.9052	0.6426	0.8856
6	0.9604	0.7468	0.8972	0.6359	0.8792
8	0.9588	0.7425	0.8913	0.6304	0.8726
10	0.9542	0.7364	0.8842	0.6158	0.8643
12	0.9435	0.7322	0.8809	0.6007	0.8507
14	0.9417	0.7069	0.8764	0.5958	0.8465
16	0.9344	0.6843	0.8715	0.5772	0.8401
18	0.9305	0.6762	0.8646	0.5620	0.8309
20	0.9275	0.6735	0.8603	0.5537	0.8216

From Table 3, when the path failure rate is 4%, the safe path rate of NFV-SDN optimization algorithm is 96.25%. SPR, NDMP-SPR, ER, and NDMP-ER algorithms' secure path rate is 75.16%, 90.52%, 64.26%, and 88.26%. When the path failure rate is set to 20%, the safe path rates of the NFV-SDN, SPR, NDMP-SPR, ER, and NDMP-ER algorithms are 92.75%, 67.35%, 86.03%, 55.37%, and 82.16%. The AM algorithm based on NFV has the highest secure path rate, but as path failure rate increases, all four algorithms' secure path rates decrease. In actual 5G core networks, the network environment is complex and ever-changing, requiring not only efficient

handling of large amounts of heterogeneous traffic but also quick response to node failures, link interruptions, and traffic surges. The NFV-SDN optimization algorithm monitors network node resources in real-time. In the event that the resource utilization rate of a node exceeds a specified threshold, the model is capable of triggering a VNF migration in a timely manner, thereby ensuring the efficient utilization of network resources. Under the condition of a path failure rate of 20%, the secure path rate of the NFV-SDN optimization algorithm is as high as 92.75%. In addition, when dealing with high-density traffic, the link load growth of NFV-SDN optimization algorithm is relatively stable, which proves its stability in real-world high load 5G networks.

4.2 Analysis and Test of Dynamic routing Model Based on MAB

This experiment is written in Python language and is conducted on a Windows 10 system using an IDEA simulation program. The learning factors are set to 0.3 and 0.6. The convergence of NFV-SDN optimization algorithm with and without link rewards is compared in Fig. 10.

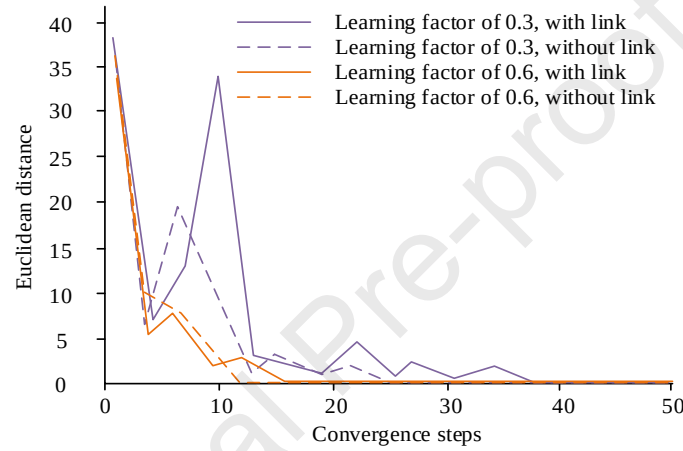


Fig. 10 Convergence with and without Link Rewards

In Fig. 10, the purple solid line represents the convergence curve with a learning factor of 0.3 and links. At this point, this model converges after 38 iterations. The purple dashed line represents the convergence curve when learning factor is 0.3 and there is no link. At this point, this model converges after 25 iterations. The orange solid line represents the convergence curve when learning factor is 0.6 with a link, where this model converges after 15 iterations. The orange dashed line represents the convergence curve when learning factor is 0.3 and there is no link. At this point, this model converges after 12 iterations. The function with parameters set converges more slowly than the function without parameters set. The NFV-SDN optimization algorithm is more complex, the convergence rate of path planning reward value is slower. Therefore, when selecting links, more iterations are required to achieve the optimal path selection. Random selection strategy, Softmax algorithm, UCB, and EG are selected. Four algorithms are tested 100 times and their total return values are compared in Fig. 11.

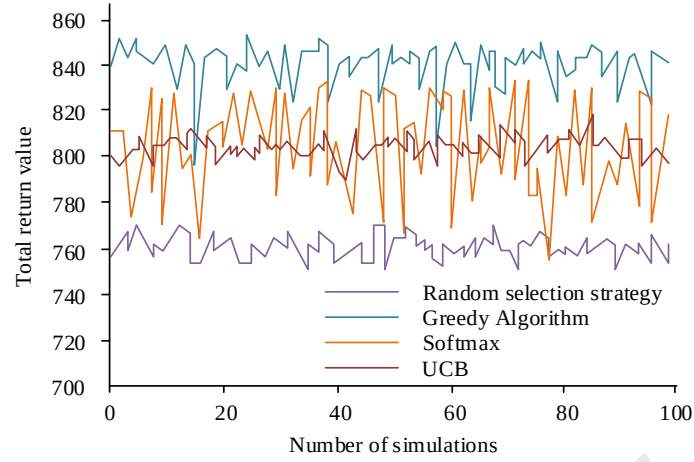


Fig. 11 Comparison Curve of Return Values of Four Algorithms

In Fig. 11, the average total return values of random selection strategy, Softmax, UCB, and EG are 768.5, 816.7, 808.6, and 842.5. UCB's fluctuation is smaller and more stable, while Softmax's average total return value is the largest. Random selection strategy's total return value is the lowest, while EG's total return value is the highest. Different learning factors and discount factors are set, and the NFV-SDN optimization algorithm is compared with the European distance of the Dijkstra algorithm. The results are shown in Table 4.

Table 4 Comparison of Euclidean Distances Between Two Algorithms

Learning factor	Discount factor	Dijkstra algorithm	NFV-SDN optimization algorithm
0.3	0.3	50.42	40.56
0.3	0.6	46.42	35.28
0.3	0.9	42.59	30.65
0.6	0.3	48.25	34.95
0.6	0.6	43.51	32.63
0.6	0.9	40.73	27.16
0.9	0.3	45.22	33.42
0.9	0.6	41.82	32.13
0.9	0.9	39.82	25.75

From Table 4, when two algorithms explore the optimal path, the learning and discount factors are larger, the NFV-SDN optimization algorithm retains the previous learning results more, and the emphasis on training experience is higher. NFV-SDN optimization algorithm is compared with Dijkstra algorithm's path transmission load. Two conditions are set for source node location and the target node to be fixed and moved in Fig. 12.

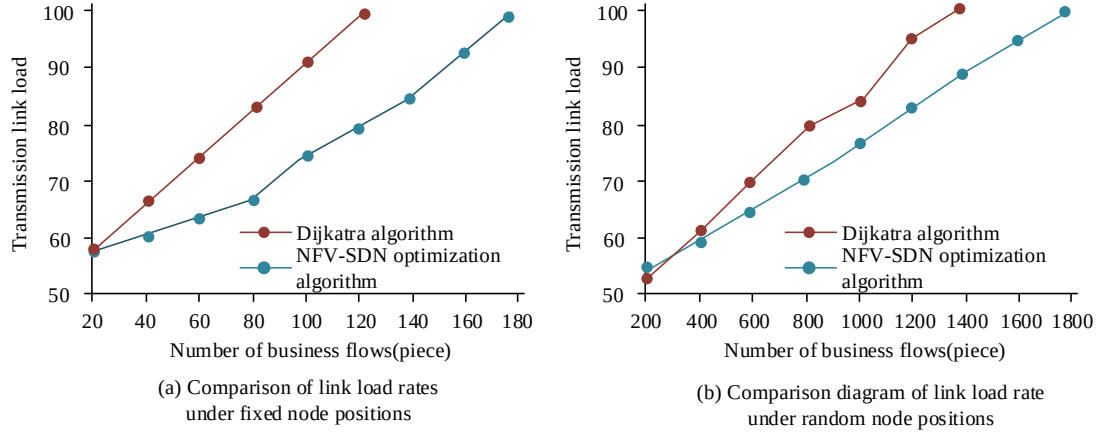


Fig. 12 Comparison of Link Delay between Two Algorithms

Fig. 12 (a) shows two algorithms' link load rates with fixed nodes. Fig. 12 (b) shows two algorithms' link load rates with mobile node. In the case of fewer traffic streams and the same number of traffic streams, Dijkstra algorithm's load rate grows rapidly. The link load rate of NfV-SDN optimization algorithm is low and the load growth is relatively stable. Therefore, the NfV-SDN optimization algorithm can effectively make link load rate reduced and link utilization improved. The link load is set at 50% and 75%, and two algorithms' link latency is compared in Fig. 13.

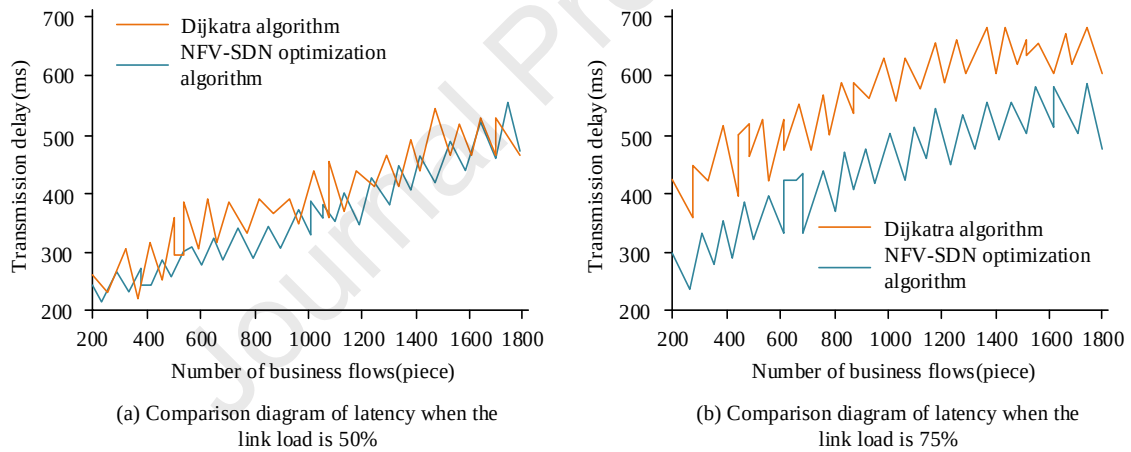


Fig. 13 Comparison of Link Delay between Two Algorithms

Fig. 13 shows two algorithms' latency comparison under different link load rates. When the link load is 50%, two algorithms' link latency is relatively close. When link load is 75%, the link delay of SDN dynamic routing model based on MAB is significantly less than Dijkstra algorithm. When network traffic flows increase, the delay of NfV-SDN optimization algorithm also increases, but it is significantly less than Dijkstra algorithm. Therefore, the NfV-SDN optimization algorithm has good performance in network traffic transportation. To further confirm the advantages and performance characteristics of the research method, advanced energy efficient offloading mechanisms, hybrid deep learning models, and flexible resource allocation optimization strategies from recent years are introduced for comparison of time complexity, computational cost, and adaptability, as shown in Table 5 [27-29].

Table 5 Comparison of Time Complexity, Computational Cost, and Adaptability

Method	NfV-SDN Optimization	Energy efficient offloading	Hybrid deep learning model	Flexible resource allocation
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	Algorithm	mechanism		optimization strategy
Time Complexity	$O(n \log n)$	$O(n^2)$	$O(n^3)$	$O(n)$
Computational Cost (ms or MB)	Moderate (350ms migration, 58% resource utilization)	Low (250ms migration, 65% resource utilization)	High (400ms migration, 60% resource utilization)	Low (300ms migration, 75% resource utilization)
Adaptability (Traffic Surge)	High: Dynamically adjusts resources	Low: Slow response to load changes	Medium: Partial load balancing	Medium: Stable under load fluctuations
Adaptability (Node Failure)	High: Path failure rate 4%-20%, safe path rate up to 96.25%	Medium: Impacted by node failure	Medium: Delayed recovery from node failures	Low: Sensitive to node failures

As shown in Table 5, the time complexity of the NFV-SDN optimization algorithm is $O(n \log n)$, which performs well in handling dynamic network environments and can effectively reduce network migration costs. However, due to the need to continuously monitor and analyze node status, the time complexity is slightly higher than that of the Flexible Resource Allocation Optimization strategy. When the NFV-SDN optimization algorithm reaches 500 nodes, its migration cost reaches 3000ms, and the resource utilization rate of the target node is 58%. Although the computational cost is moderate, compared to other algorithms, it can better balance network performance and resource utilization. The NNFV-SDN optimization algorithm exhibits high adaptability and can dynamically allocate resources based on real-time monitoring data to cope with sudden traffic. Under various traffic modes, such as random distribution and uniform distribution, it can maintain high network performance. The comprehensive performance of the research method is enhanced in dynamic network environments, particularly in the context of sudden traffic growth and node failures, which confer significant advantages. Although the research methods mainly focus on optimizing 5G core networks, they also have certain operational advantages in other types of networks. The NFV-SDN optimization algorithm is also applicable in cloud data centers. With the widespread application of virtualization technology in data centers, dynamic migration and resource optimization of VNFs have become key issues. This model can improve the utilization of server resources within the data center and reduce the impact on business during the migration process. The advantage of the MAB-based dynamic routing model in IoT networks is that it can dynamically select and optimize paths in resource constrained environments, improve data transmission efficiency, and reduce latency.

5. Conclusion

To meet the network consumption cost and function adaptation in 5G core network, this research built an AM and dynamic routing models based on NFV and MAB. The performance of NFV-SDN optimization algorithm was tested. When nodes were 200 and 500, the migration costs of NFV-SDN optimization algorithm were 1000 and 3000, respectively, with an average migration cost of 350 and 900, respectively. The resource rates were tested when target nodes were 200 and 500. The average resource occupancy rates of the destination nodes in NFV-SDN optimization algorithm were 52% and 58%, respectively. When path failure rate was 4%, the secure path rate of

NFV-SDN optimization algorithm was 96.25%. When path failure rate was set to 20%, the safe path rate of NFV-SDN optimization algorithm was 92.75%. The performance of MAB-based dynamic routing model was tested. The random selection strategy, Softmax algorithm, UCB, and EG were compared. EG's average total return value was 842.5, which was the highest compared to other algorithms. In the case of fixed and mobile nodes, the link load rate of MAB-based dynamic routing model was low and its load growth was relatively stable. When link load was 75%, the link delay of SDN dynamic routing model based on MAB was significantly less than the Dijkstra algorithm. However, as the network size expands and the number of nodes increases, the computational complexity and decision time of research methods will also increase. VNF migration will occupy bandwidth and computing resources, which may result in additional network load and latency during frequent migration. Future research could consider sinking some routing calculations to the network edge to improve response speed, and introduce migration prediction models to anticipate resource usage trends in advance and reduce unnecessary migration operations.

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Highlights:**1. Network Function Virtualization (NFV) Foundation:**

Integration of dynamic resource awareness algorithms to develop an adaptive migration model to handle increasing network demand and diverse business requirements.

2. Multi Armed Bandit (MAB) Algorithm Integration:

Implementation of a greedy algorithm based on MAB to search random actions and construct a dynamic routing model for efficient resource allocation.

3. Adaptive Migration Model Effects:

Migration costs and average migration statistics show significant improvements in resource allocation effectiveness when implementing the adaptive migration model based on NFV.

4. Algorithm Performance Metrics:

Path failure rates, safe path rates, and link load rate for fixed and mobile nodes demonstrate enhanced stability, efficiency, and reduced load growth under the dynamic routing model based on MAB algorithm.

5. Benefits of Constructed Models:

By maximizing server resource utilization, reducing cost consumption, and achieving optimal benefits, the two constructed models enable efficient network management and performance optimization.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: